

ALBEMARLE OUTSON, United Kingdom

WMO: 032380

Lat: 55.0197N

Lon: 1.8817W

Elev: 479

StdP: 14.44

Time Zone: 0.00 (EUW)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.1	27.0	20.7	15.8	26.6	23.0	17.7	29.8	37.1	46.2	33.7	46.8	5.3	290	0.628

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.8	73.1	62.9	70.0	61.5	67.2	59.8	64.6	70.8	62.8	68.1	61.1	65.6	7.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.0	84.5	67.3	60.5	80.1	65.2	59.0	75.9	63.6	29.9	70.9	28.5	68.1	27.4	65.9	71.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
27.0	23.0	19.8	DB	20.2	78.9	3.9	3.8	17.4	81.6	15.1	83.9	12.9	86.0	10.1	88.8
			WB	19.6	67.9	3.8	2.2	16.9	69.5	14.6	70.8	12.5	72.0	9.7	73.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	47.6	38.5	38.8	41.2	45.3	49.9	55.1	58.7	58.0	54.6	49.0	42.6	39.0	(6)
(7)		DBStd	8.88	5.45	5.22	5.18	4.96	4.95	4.31	4.35	3.71	4.33	4.81	5.61	6.34	(7)
(8)		HDD50	1824	358	313	276	156	62	7	0	0	9	74	228	341	(8)
(9)		HDD65	6358	823	732	738	591	469	299	203	218	314	495	672	804	(9)
(10)		CDD50	950	0	0	3	15	58	159	268	250	146	44	6	1	(10)
(11)		CDD65	10	0	0	0	0	0	1	6	2	1	0	0	0	(11)
(12)		CDH74	66	0	0	0	0	1	8	47	8	2	0	0	0	(12)
(13)		CDH80	4	0	0	0	0	0	0	3	1	0	0	0	0	(13)
(14)	Wind	WSAvg	8.5	10.7	9.1	9.8	8.8	8.5	7.2	7.0	7.3	8.0	7.9	9.1	9.3	(14)
(15)	Precipitation	PrecAvg	29.3	2.4	2.1	1.9	2.2	2.2	2.6	2.7	2.8	2.4	2.6	3.4	2.6	(15)
(16)		PrecMax	41.1	6.1	4.4	4.0	5.5	3.5	6.6	5.6	6.7	6.2	5.3	6.3	6.4	(16)
(17)		PrecMin	20.9	0.4	0.4	0.4	0.4	0.7	0.6	0.4	0.5	0.7	0.6	1.1	1.2	(17)
(18)		PrecStd	4.6	1.3	1.1	0.9	1.6	0.9	1.7	1.3	1.3	1.5	1.2	1.5	1.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.4	54.8	59.1	66.0	71.8	74.7	79.0	73.9	72.8	63.4	58.5	54.8	(19)
(20)			MCWB	50.6	49.5	51.8	56.0	61.3	63.8	64.6	64.1	62.0	58.7	55.9	51.9	(20)
(21)		2%	DB	50.8	51.0	54.1	60.3	66.2	70.4	74.4	70.0	68.0	60.5	55.0	52.5	(21)
(22)			MCWB	48.2	47.8	49.1	52.3	57.9	61.0	63.8	62.2	60.4	56.5	52.5	49.9	(22)
(23)		5%	DB	48.8	49.0	51.7	56.8	61.9	66.6	70.6	67.3	64.5	58.5	52.7	50.8	(23)
(24)			MCWB	46.4	46.1	47.6	50.3	54.3	58.7	62.3	60.1	58.4	55.4	50.4	48.4	(24)
(25)		10%	DB	46.8	46.8	49.4	54.2	58.8	63.9	67.2	65.1	62.2	56.5	50.8	48.8	(25)
(26)			MCWB	44.7	44.2	45.7	48.6	53.0	57.4	60.2	58.8	56.9	53.4	48.6	46.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.0	50.7	52.8	56.7	62.6	66.2	67.6	65.9	64.2	59.8	56.0	52.2	(27)
(28)			MCDWB	53.1	53.7	58.1	64.0	71.2	73.1	76.0	72.0	70.5	62.1	58.0	54.3	(28)
(29)		2%	WB	48.5	48.3	50.1	53.8	58.5	62.1	64.9	63.5	61.3	57.5	53.0	50.5	(29)
(30)			MCDWB	50.4	51.0	53.4	59.2	64.3	67.1	71.9	68.3	66.2	59.6	54.5	52.4	(30)
(31)		5%	WB	46.8	46.5	48.0	51.3	56.0	59.8	63.0	61.8	59.3	55.8	50.8	48.7	(31)
(32)			MCDWB	48.6	48.7	51.1	56.0	60.8	65.3	69.0	65.6	63.0	58.1	52.3	50.6	(32)
(33)		10%	WB	44.8	44.5	46.1	49.1	53.6	58.0	61.3	60.3	57.7	53.9	48.9	46.8	(33)
(34)			MCDWB	46.7	46.6	49.2	53.4	57.8	63.0	66.3	63.6	61.1	56.1	50.5	48.7	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.2	9.5	11.8	13.9	14.3	14.4	14.8	13.7	13.6	11.1	9.6	9.2	(35)
(36)			MCDBR	10.4	11.9	14.8	18.7	19.6	19.6	19.9	16.3	17.1	12.8	10.7	11.0	(36)
(37)			MCWBR	9.6	9.4	10.3	11.7	11.8	11.6	10.8	9.4	10.3	9.6	9.9	10.1	(37)
(38)		5% WB	MCDBR	10.3	11.4	13.5	17.2	18.0	17.8	18.1	14.1	15.0	11.7	10.4	11.0	(38)
(39)			MCWBR	9.8	9.4	10.3	11.5	11.5	11.4	10.6	9.1	10.0	9.7	9.9	10.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.314	0.330	0.361	0.384	0.383	0.383	0.384	0.374	0.355	0.338	0.313	0.297	(40)	
(41)		taud	2.301	2.375	2.307	2.267	2.316	2.339	2.342	2.384	2.443	2.446	2.363	2.270	(41)	
(42)		Ebn at Noon	194	238	255	266	273	273	271	266	257	232	194	171	(42)	
(43)		Edh at Noon	18	23	31	37	37	37	36	33	27	22	16	15	(43)	
(44)	All-Sky Solar Radiation	RadAvg	177	381	736	1192	1547	1543	1481	1219	868	472	226	132	(44)	
(45)		RadStd	23	41	85	118	95	134	132	85	74	45	21	13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page